

A QUANTUM CHIRPLET TRANSFORM, AUDITED: EXACT CIRCUITS, FAIR CONTROLS, AND A REAL- DEVICE TEST

Anonymous authors

Paper under double-blind review

ABSTRACT

Does a quantum-native signal representation, given every function of its classical peer, learn or reconstruct anything the classical one does not? We answer this for one representation with an audit, and the audit is the contribution. The vehicle is a quantum adaptive chirplet transform (QACT), whose atoms are exact shallow circuits. The audit has three parts: functional parity with a strong classical engine, pre-registered paired comparisons on real EEG, and a control that isolates each apparent win. Every apparent QACT advantage had a classical explanation: the transform’s free 0.5 Hz frequency axis, and a classical seed band that excludes mains interference. At equal function set the classical engine is at least as good. The quantum sampler dequantises exactly to a periodogram, computable in $O(N \log N)$. Two things survive: the dictionary is gate-native and needs no QRAM, and the matching-pursuit residual can be kept coherent by a one-ancilla reflection whose success probability equals the residual energy fraction, which we verify on circuits. On an IBM Heron processor, six-qubit selection circuits pick the exact atom in 8 of 12 cases and never fall below 0.71 of the best atom’s energy; seven-qubit circuits are marginal and long atoms fail. Downstream, the chirplet features themselves add 9–13 accuracy points over amplitude features for cross-patient seizure detection on two databases, while quantum kernels, classifiers, recurrent models and reservoirs land at parity or below. A fully controlled parity result is the baseline any quantum advantage for recorded signals has to clear.

1 INTRODUCTION

Quantum methods for recorded signals are usually judged by whether they run and whether they win. We ask the question a reviewer of any new representation would ask: *given every function of its classical peer, does a quantum-native representation learn or reconstruct anything the classical one does not, and if not, why not?* We answer it for one representation with a three-part audit: *functional parity* with a strong classical engine, so that the two are compared as equals; *pre-registered comparisons* on real EEG, with paired tests across subjects or patients; and, for each apparent win, *a control that isolates its cause*. The audit is the contribution; the transform is the vehicle.

The vehicle is the adaptive chirplet transform (ACT) of Mann & Haykin (1991; 1992; 1995). It writes a signal as a few Gaussian-windowed linear chirps chosen greedily by matching pursuit (Mallat & Zhang, 1993; Bultan, 1999). The chirp rate, which a stationary spectrum cannot report, makes it informative for drifting EEG transients. The quantum version rests on one observation: a chirp’s quadratic phase is exactly 2-local in the binary encoding of time, so an atom is an $O(n^2)$ -gate circuit on $n = \log_2 N$ qubits. One inverse quantum Fourier transform (QFT) then returns an atom’s whole frequency axis, measurement samples atoms with the probability matching pursuit maximises, and the parameter-shift rule refines every phase parameter exactly (§3). Earlier quantum matching-pursuit proposals assume quantum random-access memory (QRAM) for signal and dictionary, and have not been run on data (Bellante & Zanero, 2022; Bellante et al., 2025).

The obvious objection. A reader’s first objection is that this sampler is just a periodogram. It is. For each envelope and chirp rate, the sampled distribution is the squared magnitude of one FFT,

computable classically in $O(N \log N)$, and a classical argmax over it is the transform with infinitely many shots (Proposition 1). Two pieces of quantum structure survive. The dictionary is gate-native, so a superposition over atoms costs controlled gates, not a QRAM holding the dictionary. And the matching-pursuit residual can be kept coherent, with success probability equal to the fraction of energy still unexplained (Proposition 2). Neither is a speedup at EEG sizes (§4). Why audit it, then? Because apparent quantum wins on real data are easy to produce and hard to dissolve: this transform produced three apparent wins, each with a classical cause that only a targeted control revealed. And a representation is where a classical baseline can be matched exactly, since the classical peer is a program, not a model class. A controlled parity result there is the bar a claimed advantage has to clear.

What the controls found. QACT first appeared to win on denoising against ICA-cleaned ground truth (17 of 18 held-out subjects, $p < 10^{-4}$). No bug produced the win. Its cause was the transform’s frequency axis (every 0.5 Hz QFT bin from 0.5 to 45 Hz, against the shipped classical grid’s 1 Hz steps over 1–40 Hz), which the classical engine can copy. A hardware-motivated refiner then appeared to beat the frequency-matched classical engine. Giving the classical engine the same refiner does *not* close that gap; extending its seed grid past 45 Hz, to the 50 Hz mains in the injected artifacts, does. On public data the shipped grid failed on windows dominated by 60 Hz mains, so QACT looked better in the mean but not the median. Each effect is a legitimate engineering finding; none is a quantum one. Learning has one positive result (Table 4): chirplet features add 9–13 accuracy points over amplitude features for cross-patient seizure detection, on two databases and two classifiers. Swapping in QACT does not raise that number, and quantum kernels, variational classifiers, quantum recurrent models and quantum reservoirs land at parity or below.

Contributions. (1) A pre-registered audit on real EEG (§5): denoising with ground truth on 26 subjects, seizure detection on 37 patients from two databases, and per-person forecasting on 983 hours, with the controls that dissolved each apparent advantage (three run for this paper). (2) A quantum ACT with exact circuits for the atom, the selection step and the refinement gradient, at full functional parity with a strong classical engine (§3); the post-selection-conditioned distribution a device reports *is* the classical normalised criterion, which fixed a 70% deficit. (3) The exact dequantisation of the sampler, and what survives it (§4): a gate-native dictionary with no QRAM, and a coherent residual with success probability equal to the residual energy fraction, verified on circuits to 10^{-9} . (4) The first execution of the transform’s selection step on a quantum processor (§6): a short atom cut from 2,027 to about 140 CZ gates, selecting the exact atom at six qubits, degrading at seven, and failing for long atoms. Code, per-window records and the pre-registered comparisons are released.

2 BACKGROUND

The transform. Let $x \in \mathbb{R}^N$, $N = 2^n$. A chirplet atom in the sample domain is

$$\psi_\theta(t) = \frac{1}{\|\psi_\theta\|} e^{-(t-t_c)^2/2\Delta t^2} \exp\left(\frac{i2\pi}{N} [c(t-t_c)^2 + f_c(t-t_c)]\right), \quad \theta = (t_c, f_c, \Delta t, c), \quad (1)$$

with f_c in cycles per window and c in cycles per window per sample. Matching pursuit picks the atom maximising $|\langle \psi_\theta | r \rangle|^2$ against the residual r , refines θ off-grid, subtracts, and repeats; orthogonal matching pursuit (OMP) re-fits all coefficients jointly (Pati et al., 1993). The classical engine used throughout is a batched GPU implementation of the ACT (Vicol, 2026); residual-projection matching pursuit and hybrid GPU acceleration for the ACT are developed by Kumar & Mann (2026). The engine selects by the least-squares energy of the cosine/sine quadrature pair, refines with sixty steps of Adam on the same objective, re-fits by OMP, and stops per window by rule. A two-width asymmetric envelope (Vicol, 2026) is available as a variant.

Seed grids. All classical seed grids have 17 centres, 6 widths and 5 chirp rates and differ only in frequency; we name the grid wherever it matters. The *shipped grid* (the engine as released, used unless another is named) has 40 frequencies at 1 Hz, 1–40 Hz: $17 \times 40 \times 6 \times 5 = 20,400$ atoms. The *EEGMAT 0.5 Hz grid* (EEGMAT grid for short), used for denoising, matches QACT’s band and spacing (every QFT bin of a 512-sample window at 256 Hz): 90 frequencies, 0.5–45 Hz, $17 \times 90 \times 6 \times 5 = 45,900$ atoms. The *public 0.5 Hz grid*, used on the public motor-imagery windows,

keeps the shipped band at 0.5 Hz spacing (80 frequencies, 1–40.5 Hz, 40,800 atoms). The seed-band controls extend the EEGMAT grid (62,730 atoms) or the shipped grid (39,780 atoms) to 78 Hz in 1 Hz steps.

Quantum learning on EEG. Quantum kernels (Havlíček et al., 2019; Huang et al., 2021), re-uploading classifiers (Pérez-Salinas et al., 2020), quantum recurrent models (Chen et al., 2020) and quantum reservoirs (Fujii & Nakajima, 2017; Martínez-Peña et al., 2021) have all been applied to biosignals, usually on features a classical transform produced. Bowles et al. (2024) showed that most reported quantum-versus-classical wins in learning do not survive matched baselines; Wolff et al. (2026) found a quantum reservoir that wins on benchmarks and loses on EEG. Our audit follows that discipline and moves it one step upstream, into the representation.

3 THE QUANTUM ADAPTIVE CHIRPLET TRANSFORM

3.1 THE ATOM IS A CIRCUIT

Expanding $c(t - t_c)^2 + f_c(t - t_c) = ct^2 + (f_c - 2ct_c)t + \text{const}$ shows that a shifted chirp is the same quadratic phase plus a linear phase at the *effective frequency* $f = f_c - 2ct_c$; the centre time enters only through the envelope. With $t = \sum_k b_k 2^k$ and $b_k^2 = b_k$,

$$t^2 = \sum_k b_k 2^{2k} + 2 \sum_{j < k} b_j b_k 2^{j+k} \Rightarrow e^{i2\pi ct^2/N} = \prod_k P_k\left(\frac{2\pi c 2^{2k}}{N}\right) \prod_{j < k} CP_{jk}\left(\frac{2\pi c 2^{j+k+1}}{N}\right), \quad (2)$$

a product of n phase gates and $n(n-1)/2$ controlled-phase gates, exactly.¹ The linear phase is n single-qubit gates. A cubic phase, an atom whose chirp rate itself evolves, is 3-local by the same argument (Table 1). The Gaussian envelope is a real diagonal filter, realised either by a multiplexed rotation onto an ancilla with post-selection or, as we do on hardware, by folding it into state preparation (§6).

3.2 SELECTION BY ONE TRANSFORM AND ONE MEASUREMENT

Prepare $|s\rangle$ with $s = e \odot r / \|e \odot r\|$ for envelope $e = e_{t_c, \Delta t}$, apply the conjugate chirp for rate c and an inverse QFT, and measure the register. The outcome k has probability

$$P_{e,c}(k) = \frac{1}{N} \left| \sum_t s(t) e^{-i2\pi ct^2/N} e^{-i2\pi kt/N} \right|^2. \quad (3)$$

Rescaled by the classical factor $N\|e \odot r\|^2/\|e\|^2$, this equals $|\langle \psi_{e,c,k} | r \rangle|^2/\|\psi\|^2$: the energy of atom $(t_c, \Delta t, c, k)$ captured from r , normalised by the atom’s norm. One circuit therefore covers every frequency of the dictionary for one $(t_c, \Delta t, c)$ triple. The remaining triples are enumerated (one circuit each) or, in principle, superposed in index registers and searched by quantum maximum finding (Dürr & Høyer, 1996) (§4). Sampling the outcome makes selection stochastic. The `argmax` over the identical dictionary is the control that isolates measurement from formulation, and `shots` interpolates between the two.

That normalisation was the source of the transform’s first failure. The raw histogram is the *joint* probability that the envelope filter succeeds and frequency k is read, so it over-ranks wide, high-norm envelopes. A device keeps only post-selected shots, so what it reports is the *conditional* distribution. Dividing by the filter’s success probability $\|\psi\|^2$ makes the hardware observable, the refiner’s objective and the classical least-squares criterion the same quantity. This one change moved QACT from a reconstruction error of 0.307 to 0.169 on planted chirplets and, on real EEG denoising, from 70% worse than the classical engine to statistically indistinguishable from it (§5.1).

3.3 EXACT REFINEMENT BY THE PARAMETER-SHIFT RULE

The captured energy $E(\theta) = |\langle \psi_\theta | r \rangle|^2/\|\psi_\theta\|^2$ depends on every phase-gate angle a through a factor e^{ia} multiplying part of the overlap. So $E = A + B \cos a + C \sin a$, and $\partial E/\partial a = [E(a +$

¹Verified against the analytic diagonal and against a statevector simulation (Javadi-Abhari et al., 2024) to 9×10^{-13} and 4×10^{-14} .

Table 1: Functional parity between the classical engine and QACT. “Cost” is what the function costs on a device beyond the selection circuit.

classical engine	QACT	cost on a device
dictionary search, least-squares energy	one circuit per $(t_c, \Delta t, c)$; histogram conditioned on post-selection	shots
Adam refinement of $(t_c, f_c, \log \Delta t, c)$	parameter shift for phases, central differences for envelope or three-point coordinate search	112 evaluations/step 8 evaluations/step
OMP joint refit	Gram of atom overlaps; parameters known, so classical	none
two-width asymmetric envelope	fourth envelope grid axis, refinable	none
stopping rule (active mask)	same semantics	none
–	exact frequency update (periodogram read-out)	one circuit
–	backfit: cyclic re-refinement of each atom	refinement $\times$ atoms
–	cubic phase (3-local) and skew envelope, refinable	129 gates at $n = 9$

$\pi/2) - E(a - \pi/2)]/2$ exactly (Mitarai et al., 2018; Schuld et al., 2019). The chain rule maps the $n + n(n - 1)/2$ angles onto f and c with the weights of eq. equation 2. The envelope parameters are not gate angles and use central differences, as a device would. Closed-form and genuinely shifted evaluations agree to 2.8×10^{-9} ; each step costs 112 circuit evaluations, and four steps cut reconstruction error on planted signals by about 25%. A cheaper rule, used on hardware, is a three-point coordinate search (two evaluations per coordinate, keep the best of three) at 8 evaluations per step. It scores *better* than the gradient on denoising, an effect we later have to explain (§5.1).

3.4 FUNCTIONAL PARITY

A transform that only selects atoms is not the classical engine. Table 1 lists every function of the classical engine and its quantum-native counterpart. Each is opt-in, and with all of them off the transform reproduces the pre-parity reference feature matrix bit for bit (2.5 million entries). The OMP refit costs no circuits: its normal matrix is the Gram matrix of atom overlaps $\langle \psi_i | \psi_j \rangle$, a swap-test observable in principle, but it depends only on atom parameters, so it and the right-hand side are classical. The exact frequency update follows from eq. equation 3: $E(f)$ is the periodogram, so the optimal f is read off one transform instead of gradient-stepped toward. Two functions have no counterpart in the classical engine we compare against (the exact frequency update and backfit), so the control study of §5.2 adds backfit to the classical engine as well.

4 WHAT THE QUANTUM STRUCTURE BUYS, EXACTLY

Proposition 1 (Exact sampler and its dequantisation). *For any envelope e , chirp rate c and residual r , the outcome distribution of the selection circuit is eq. equation 3, and $N \|e \odot r\|^2 \|e\|^{-2} P_{e,c}(k) = |\langle \psi_{e,c,k} | r \rangle|^2 / \|\psi_{e,c,k}\|^2$ for every k . All N values are computed classically by one fast Fourier transform of $e \odot r \odot e^{-i2\pi c t^2/N}$ in $O(N \log N)$ operations.*

The proof is the definition of the QFT (Appendix A). The consequence is that an *enumerated* QACT with S shots per triple costs $\Theta(M' S N)$ gate-time per iteration (state preparation of an arbitrary N -vector needs $\Theta(N)$ gates without QRAM (Möttönen et al., 2005; Shende et al., 2006)) against $\Theta(M' N \log N)$ classical operations. A classical program that computes eq. equation 3 by FFT and takes the argmax is exactly the transform with infinitely many shots. We ran that program. On the GPU it decomposes a window in 0.088 ms (batched), faster than the classical engine (0.672 ms) because it uses four cheap refinement steps instead of sixty, at a worse fit (0.58 versus 0.47). It is a classical algorithm that computes what the circuit would.

Two pieces of quantum structure are not dequantised by Proposition 1. The first is that the dictionary is *gate-native*: an atom is $O(n^2)$ gates plus a Gaussian preparation (Grover & Rudolph, 2002). A superposition over the M' triples in index registers therefore costs controlled versions of those gates, not a QRAM of $M' N$ amplitudes as quantum matching pursuit assumes (Bellante & Zanero, 2022; Jaques & Rattew, 2025). Quantum maximum finding over the index registers then selects the best triple in $O(\sqrt{M'})$ evaluations (Dürr & Høyer, 1996; Brassard et al., 2002). The second is the residual.

Proposition 2 (Coherent residual). *Let $P_k = I - |\psi_k\rangle\langle\psi_k|$ and $R_k = I - 2|\psi_k\rangle\langle\psi_k| = U_k(I - 2|0\rangle\langle 0|)U_k^\dagger$, where $U_k|0\rangle = |\psi_k\rangle$ is the atom’s preparation circuit. With one ancilla,*

$(H \otimes I) c\text{-}R_k (H \otimes I)$ applied to $|0\rangle|r\rangle$ and post-selected on the ancilla reading 0 yields $P_k|r\rangle$ with probability $\|P_k r\|^2/\|r\|^2$. Chaining K such steps on the loaded signal $|x\rangle$ yields exactly the classical matching-pursuit residual r_K , with success probability $\|r_K\|^2/\|x\|^2$, the fraction of the signal’s energy the K atoms leave unexplained.

The proof is two lines (Appendix A). We verified it on a real EEG window with $N = 64$ and $K = 4$. The chained projectors match the classical residual to $1e - 16$ in a numpy statevector, and to $8e - 10$ in a circuit built from state preparation, phase gates and a multi-controlled phase, with success probability 0.307 equal to $\|r_K\|^2/\|x\|^2$ to six digits. One controlled reflection costs 254 two-qubit gates at $n = 6$ on an all-to-all CZ basis, dominated by the two envelope preparations. On the 320 public EEG windows of §5.2, the success probability is 0.43, 0.29, 0.21 and 0.16 at 8, 16, 24 and 32 atoms (Fig. 3, Appendix A). The algorithm slows down precisely as it succeeds, and amplitude amplification keeps the repetition count between two and three.

The coherent residual removes the classical computation of r_k between iterations, not the $\Theta(N)$ load of x , which every Grover repetition repeats. Without QRAM the superposed variant therefore costs $O(\sqrt{M'} \varepsilon^{-1} (N + Kn^2))$ gate-time per iteration for amplitude resolution ε , against $O(M' N \log N)$ classically. The gain is polynomial in the dictionary size only, and the clock-speed and shot costs of §6 eat it at the dictionary sizes EEG needs ($M' \leq 5,000$). The dequantisation literature (Aaronson, 2015; Tang, 2019; Chia et al., 2020; Tang, 2022) predicts exactly this split (the QRAM regime is in Appendix A), and the experiments below are consistent with it.

5 EXPERIMENTS ON REAL EEG

Protocol. Every comparison was pre-registered with its decision rule before it was run: subject-wise or patient-wise folds, paired Wilcoxon signed-rank tests across subjects or patients, and Bonferroni correction over the tests run on a dataset. The two frequency-matched denoising arms were added after the parity result was seen, and were named as the deciding controls before they ran. Negative results are reported in place. Seed grids are named as in §2. Datasets: EEGMAT (Zyma et al., 2019), 36 subjects, ICA-cleaned by its authors, 500 Hz resampled to 256 Hz; CHB-MIT (Shoeb, 2009), 23 cases of 22 paediatric patients, 198 seizures, 983 h; the Siena scalp EEG database (Detti, 2020), 14 patients, 45 seizures, used once as a confirmation set; and, for the controls added in this paper, the PhysioNet EEG motor movement/imagery baseline runs (Schalk et al., 2004; Goldberger et al., 2000), 20 subjects, read with MNE (Gramfort et al., 2013). All data are public (Goldberger et al., 2000).

5.1 DENOISING AGAINST GROUND TRUTH

Reconstruction error cannot measure denoising, because leaving noise in the residual is what a denoiser should do. EEGMAT is ICA-cleaned, so it serves as a clean reference. Known artifacts (blinks, horizontal eye movement, EMG, the subject’s own ECG, electrode pops, mains, drift, with per-channel propagation) are injected. Each engine decomposes the corrupted signal, atoms flagged as artifact by an amplitude-gated rule are dropped, and the rebuilt signal is scored against the clean one by a time-domain fidelity plus per-band power error in dB. Thresholds are tuned on eight subjects and frozen; 18 others are held out (26 of the 36 subjects are used). Table 2 gives the arms.

The reading changed twice. Before parity, QACT was significantly worse on every subject. The discriminating band was delta: its slow atoms matched the waveform less precisely, and subtracting them removed real EEG. After parity, every QACT arm beat the classical engine on 17 or 18 of 18 subjects. Two controls left that standing: holding QACT to the classical function set, and giving the classical engine its own asymmetric envelope. The third dismantled it. QACT’s dictionary covers every QFT bin from 0.5 to 45 Hz, because one transform returns the whole axis; the shipped grid steps at 1 Hz over 1–40 Hz. On the EEGMAT 0.5 Hz grid, which has *more* atoms than QACT (45,900 versus 22,784), the classical engine scores 1.257, and the paired test against QACT at parity finds no difference ($p = 0.15$). The arm holding QACT to the classical function set is then significantly worse. The frequency axis is a real structural property of the transform, worth more than any engine

Table 2: EEGMAT denoising, 18 held-out subjects: score (mean over subjects, lower is better; time fidelity plus per-band dB error), relative RMSE and delta-band error (medians over subjects), and the paired comparison (subjects better, of 18; Wilcoxon p). Grids as in §2; “vs grid” is against the classical EEGMAT grid (1.257), “vs hw” against QACT with the hardware refiner (1.213). Bonferroni $\alpha = 0.0056$ (nine tests) for the first block, 0.0125 (four) for the second and for the two arms run for this paper; the hardware-refiner test clears both.

engine	score	rRMSE	δ err (dB)	paired test
no cleaning	3.487	—	—	
classical engine (shipped grid)	1.439	0.428	0.73	reference
QACT before parity (selection only)	2.461	0.535	2.07	worse on 18/18, $p < 10^{-4}$
QACT at parity	1.272	0.398	0.66	better on 17/18, $p < 10^{-4}$
QACT restricted to the classical function set	1.299	0.430	0.72	better on 17/18, $p < 10^{-4}$
QACT at parity + asymmetric envelope	1.159	0.402	0.57	better on 18/18, $p < 10^{-4}$
<i>controls: the classical engine given what QACT has</i>				
classical, shipped grid + asymmetric envelope	1.375	0.417	0.71	
classical, EEGMAT 0.5 Hz grid (45,900 atoms)	1.257	0.423	0.62	
classical, EEGMAT grid + asymmetric envelope	1.160	0.408	0.60	
<i>the remaining gap: QACT’s hardware refiner, and two controls run for this paper</i>				
QACT at parity + hardware coordinate refiner	1.213	0.392	0.57	vs grid: better on 14/18, $p = 0.0007$
classical, EEGMAT grid + coordinate refiner	1.279	0.422	0.61	vs hw: better on 1/18, $p = 4 \times 10^{-5}$ vs grid: better on 5/18, $p = 0.05$
classical, EEGMAT grid extended to 78 Hz	1.223	0.380	0.61	vs hw: better on 5/18, $p = 0.37$: no difference vs grid: better on 18/18, $p = 8 \times 10^{-6}$
QACT at parity vs classical, EEGMAT grid				better on 6/18, $p = 0.15$: no difference
QACT restricted vs classical, EEGMAT grid				better on 4/18, $p = 0.008$: classical better
QACT parity + asym vs classical, EEGMAT + asym				better on 11/18, $p = 1.00$: no difference

function measured here.² It is also a constant factor, which the classical engine buys with a $2.25\times$ larger dictionary.

One gap remained. With the hardware refiner of §6 (three-point coordinate search), QACT scores 1.213, better than the classical engine on the EEGMAT grid (1.257) on 14 of 18 subjects ($p = 0.0007$). The obvious explanation, the refinement algorithm, is wrong. Given the same coordinate search, the classical engine scores 1.279, *worse* than QACT with the hardware refiner on 17 of 18 subjects ($p = 4 \times 10^{-5}$). Given the EEGMAT grid extended from 45 to 78 Hz, so that it can seed the 50 Hz mains the artifact injection contains, it scores 1.223. That is indistinguishable from QACT (better on 5/18, $p = 0.37$), and better than the unextended EEGMAT grid on 18 of 18 subjects ($p = 8 \times 10^{-6}$). QACT’s exact frequency update reads the periodogram peak wherever it is; from the EEGMAT grid’s top seed at 45 Hz, sixty Adam steps reach only 45–47 Hz. The seed band, not the refiner, is the explanation, and every denoising advantage of the transform now has one.

5.2 THE REFINEMENT AND DICTIONARY CONTROLS

The denoising controls isolate the seed band on one dataset and one task. To separate refiner, dictionary and band on public data, with no artifact rule in the way, we run the same factors on 320 public windows (20 subjects, two baseline runs, eight channels, 512 samples at 160 Hz, zero-mean, unit norm). The score is the relative residual norm after OMP at 6 and 12 atoms (Table 3; error against time is plotted in Fig. 4, Appendix B). The engines are the classical engine as shipped (shipped grid, Adam, 60 steps), with QACT’s budget (Adam, 4 steps) and with QACT’s coordinate rule (4 sweeps, step sizes in physical units); the public 0.5 Hz grid with Adam and with the coordinate rule; and QACT with parameter shift, with the coordinate rule, and in its full parity configuration (coordinate rule, exact frequency update, one backfit pass).

Three things follow. First, at equal refinement rule the classical engine is better than QACT. With the same coordinate search it is -3.6% below QACT at order 12, better on 264 of 320 windows ($p = 2 \times 10^{-32}$). The coordinate search itself costs the classical engine $+1.8\%$ against sixty Adam steps, at a tenth of the time, consistent with the denoising arm above. Second, finer spacing alone

²The EEGMAT grid differs from the shipped grid in band as well as spacing, so this gain combines the two; no arm separates them, and neither is quantum. On the public windows, finer spacing alone changes nothing (§5.2).

Table 3: Refinement and dictionary controls on 320 public EEG windows: mean relative residual after OMP, change versus the classical engine as shipped, windows better than it, paired Wilcoxon p , and GPU time per window (batch of 320). Order 6 / order 12.

engine	err@6	vs ref	err@12	vs ref	ms/window@12
classical, shipped grid, Adam 60 steps (reference)	0.588	—	0.486	—	43.5
classical, shipped grid, Adam 4 steps	0.631	+7.4%*	0.532	+9.5%*	3.1
classical, shipped grid, coordinate search, 4 sweeps	0.607	+3.4%*	0.494	+1.8%*	6.0
classical, public 0.5 Hz grid, Adam 60	0.588	+0.1% ($p = 0.27$)	0.485	-0.2% ($p = 0.15$)	30.5
classical, public 0.5 Hz grid, coordinate 4	0.604	+2.7%*	0.490	+0.9%*	6.4
QACT, parameter shift, 4 steps	0.638	+8.6%*	0.520	+7.1%*	3.9
QACT, coordinate search, 4 sweeps	0.631	+7.5%*	0.513	+5.6%*	4.8
QACT parity (+ exact f , backfit 1)	0.591	+0.6% ($p = 2 \times 10^{-10}$)	0.455	-6.3% ($p = 0.05$)	8.6

*: $p < 10^{-9}$ paired. Order-12 medians and the per-window records are in Appendix B.

Table 4: Downstream comparisons, all pre-registered, patient- or subject-wise folds. “Gap” is the best quantum model minus the best classical model unless stated; p is the paired Wilcoxon across patients or subjects.

task, data	comparison	gap	verdict
seizure, CHB-MIT (23 patients)	chirplet + amplitude vs amplitude only, logistic regression	+8.6 pts	$p = 0.0011$, 20/23 patients
seizure, Siena (14, held out)	same, two classifiers	+9.4, +13.3 pts	$p = 0.005$, 0.004: replicates
seizure, CHB-MIT	QACT (parity engine, sampled) vs classical features, all features + regularisation	-4.9, -4.5, -7.2 pts	$p = 0.0006$, 0.0027, $< 10^{-4}$: classical better
seizure, Siena	QACT (parity engine) vs classical features; with amplitude	-2.8, -1.8; -3.6, -5.3 pts	$p = 0.0012$ for the last: classical better
seizure, CHB-MIT	best quantum classifier vs classical RBF SVM	-0.5, -0.8 pts	$p = 0.19$, 0.61: parity
rest vs arithmetic, EEGMAT (36)	QSVC / projected kernel / re-uploading / hybrid vs logreg	-1.3 to -8.8 pts	never ahead
forecasting, CHB-MIT (23 people)	quantum LSTM vs classical twin (skill)	-0.005	$p = 0.086$: none
forecasting, CHB-MIT	quantum reservoir vs echo-state network	-0.007	$p = 10^{-4}$, ESN wins 20/23
forecasting, CHB-MIT	quantum reservoir vs its zero-coupling twin, 2 s steps (60 s steps)	+0.024 (-0.004)	$p < 10^{-4}$, 23/23 ($p = 0.23$): resolution-dependent

(the public 0.5 Hz grid, same band as the shipped grid) changes nothing on this metric at these orders (-0.2% at order 12, $p = 0.15$). The EEGMAT grid’s denoising gain in §5.1, which combines finer spacing with a wider band, is a property of the denoising task, whose mechanism we did not isolate. Third, QACT at parity is -6.3% *better* than the reference in the mean at order 12, but better on only 104 of 320 windows, with a median difference of +0.019. The mean is carried by 14 windows on which the classical engine leaves more than 90% of the energy. Those windows are dominated by 60 Hz mains interference (84% of their energy above 45 Hz, against 11% for the rest). The shipped grid stops at 40 Hz and cannot seed it; QACT’s exact frequency update reads the periodogram peak wherever it is. Extending the shipped grid to 78 Hz leaves 0 such windows, and the classical engine is then -12.6% relative to QACT parity, better on 289 of 320 ($p = 2 \times 10^{-39}$; with backfit added, Appendix B). Every remaining difference is a band, dictionary or refiner choice the classical engine can make.

5.3 DOWNSTREAM: DOES THE REPRESENTATION CHANGE WHAT IS LEARNED?

The features are 13 energy-weighted aggregates per channel (band fractions, oscillatory versus transient split, spectral centroid and spread, sweep, duration, reconstruction error), invariant to the greedy order, plus per-channel log-RMS and log line length (Esteller et al., 2001). Table 4 summarises the pre-registered comparisons; Appendix C has the full tables.

The chirplet features are the one solid positive result. They add 9–13 accuracy points beyond amplitude features for cross-patient seizure detection, significant on two databases and two classifiers (absolute accuracies in Appendix C). The quantum transform does not improve that number. Under univariate top- k selection QACT features led by 1.2 to 3.0 points, never significantly; under all features with regularisation the classical features led. With the parity engine, measurement-sampled QACT is 4.5 to 7.2 points behind on CHB-MIT and 1.8 to 5.3 behind on Siena, because sampling atoms by measurement costs 4–5 points against the argmax on the same engine. Atom-level improvements that are real produced no downstream gain: reconstruction quality and classification value were close to unrelated, three times over (Appendix C). Among the quantum learners, one effect survived its own control, and only at fine time resolution. At 2 s steps entanglement contributes to the quantum reservoir’s forecasting skill for all 23 people; at 60 s steps it does not; and the reservoir loses to a classical echo-state network of the same readout size at every resolution.

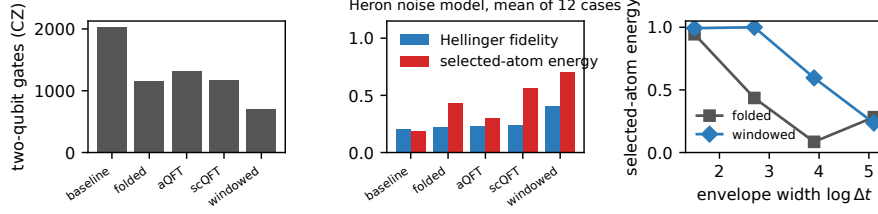


Figure 1: Five constructions of the same selection distribution on real EEG windows (12 window $\times$ width cases), transpiled to an IBM Heron target: two-qubit gates, and Hellinger fidelity plus selected-atom energy under the device’s calibrated noise model (left, centre); selected-atom energy by envelope width for the folded and windowed constructions (right).

6 ON HARDWARE

On a device the bill is circuits $\times$ shots $\times$ two-qubit gates, and the construction as drafted was expensive on all three. The envelope as a 2^n -way multiplexed rotation with post-selection discarded 43–97% of shots on real EEG, and a nine-qubit circuit transpiled to 2,027 CZ gates on a heavy-hex IBM Heron target. Three changes, each exact, reduce it (Fig. 1). *Folding* the envelope into state preparation loads $e \odot r$ directly: no ancilla, no multiplexor, every shot kept (1,156 CZ). The *semiclassical* inverse QFT of Griffiths & Niu (1996) measures each qubit as soon as it is final and replaces every controlled phase by a classically controlled single-qubit phase, so the QFT contains no two-qubit gates (1,181 CZ; a dynamic circuit). *Window-local registers* use shift covariance, $c(u + t_0)^2 = cu^2 + 2ct_0u + \text{const}$: a Gaussian atom needs a register only as large as its $\pm 4\sigma$ support, the offset folds into one linear phase, and the smaller QFT returns the original distribution exactly on every 2^{n-m} -th bin. A short atom then needs 6 qubits and about 140 CZ gates. Under the Heron noise model its selection survives (selected-atom energy 0.71 on average over widths, 0.99–1.00 for the two short widths) where the baseline never did (0.19). Long atoms ($\log \Delta t \geq 3.9$) span the window, cannot be shrunk, and lose their selection under noise; what remains for them is state preparation, about 87% of the two-qubit budget (Holmes & Matsuura, 2020; Moosa et al., 2023). Branch-and-bound over envelopes and the three-point refiner cut circuit executions per atom about 80-fold (Appendix E).

Real device. We executed the windowed construction on `ibm_marrakesh`, a 156-qubit Heron processor, for 26 circuits: six public EEG windows (three subjects, two channels) at two short widths, each in the semiclassical (dynamic) form and in a unitary-QFT form that admits gate twirling, plus one long-atom case in both forms. Each ran 4,000 shots with dynamical decoupling (jobs `dar9mejojks738nr40g`, `dar9sp3g95ks73ee22j0`, 36 s of processor time). The classical certificate for every circuit is the exact distribution of Proposition 1. The result splits by register size. The 12 six-qubit circuits (158 CZ on average) selected the exact in-band atom in 8 cases, and the atom they did select carried 0.96 of the best atom’s energy on average, never less than 0.71: selection survives (Fig. 2). The 12 seven-qubit circuits (330 CZ) are marginal, with the exact atom in 5 cases, a mean of 0.69, and 4 outright losses. The two nine-qubit long-atom circuits (1166 CZ) reached fidelity 0.30 and selected-atom energy 0.26: selection is lost, as the noise model predicted. The calibrated noise model is optimistic on this device by about 0.15–0.2 in fidelity (Appendix E).

Speed. Priced on the same job (four EEGMAT windows, six atoms, OMP; Table 6, Appendix E), the optimised circuits would take 758 s per window on a Heron processor, and 8.0 s with three published throughput techniques. That is still about $140\times$ slower than the classical engine on a CPU and $12,000\times$ slower than the batched GPU engine. Per shot, the $250\mu\text{s}$ repetition delay dwarfs the $16\text{--}96\mu\text{s}$ circuit, and each circuit needs thousands of shots to estimate what an FFT computes exactly in microseconds.

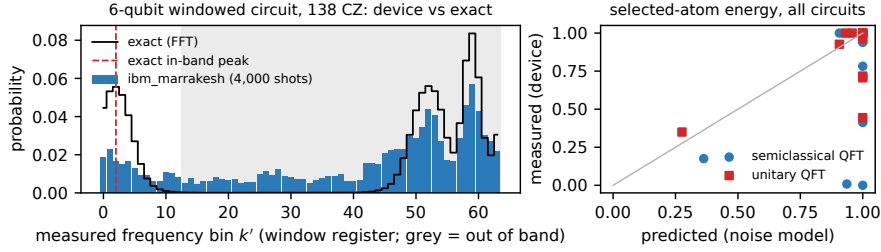


Figure 2: Left: device histogram of a 6-qubit windowed selection circuit on `ibm_marrakesh` against the exact distribution; grey bins are out of band (mirror frequencies) and are masked before selection. Right: selected-atom energy predicted by the calibrated noise model against the value measured on the device, all circuits.

7 DISCUSSION

What survives. Three things in QACT are genuinely quantum-native, and none is an advantage. The frequency axis is free, a constant factor the classical engine buys with a larger grid. The dictionary is gate-native, which removes the dictionary half of the QRAM assumption; the signal half remains, and the coherent residual costs a success probability equal to the unexplained energy. And the shortest atoms select correctly on today’s hardware, a statement about circuit design under noise, not speed. Every other apparent win had a classical explanation found by a control; one (17 of 18 subjects, $p < 10^{-4}$) fell only to the third control. This is the matched-baseline discipline Bowles et al. (2024) and Schuld & Killoran (2022) ask of quantum learning, applied where it is easiest to satisfy.

A search that ended negative. After parity, the study searched for any quantum chirplet transform that beats the strongest classical baseline, including a classical port of the same idea, in speed at equal accuracy or accuracy at equal time, a criterion fixed in advance. None did. Hybrid routing, error correction, a powered 60-window test, fault-tolerant coherent search, kernels on atoms, small-data regimes and entangled sensing all land at parity or below, or cross over only at unreachable scales (Appendix D).

Why no advantage should be expected here. Recorded signals must be loaded into every circuit for every shot, which is Aaronson’s fine print (Aaronson, 2015). Once the classical side gets equivalent access, the advantage is at most polynomial (Tang, 2019; Chia et al., 2020), and Proposition 1 makes this exact for the transform. The published exponential advantages for signals concern signals that reach a qubit directly (Huang et al., 2022; Kannan et al., 2026); an EEG amplifier does not.

Limitations. The audit covers one transform family, three EEG databases and one device. The hardware runs demonstrate selection for individual circuits, not a full decomposition on a QPU. The coherent-residual construction is verified at $n = 6$ with an unoptimised Gaussian preparation. The learners are those a practitioner would try. Nothing here rules out an advantage for a quantum-native signal.

8 CONCLUSION

For the adaptive chirplet transform, the answer to our question is no: given every function of its classical peer, the quantum-native representation learned or reconstructed nothing the classical one did not. It can be built exactly, refined exactly, brought to full functional parity with a strong classical engine, and run for the shortest atoms on a quantum processor. Under pre-registered comparisons with paired tests on real EEG it is its classical peer’s equal and beats nothing; each apparent advantage was a dictionary or seed-band choice the classical engine can copy, and three of the controls that show it are new here. A fully controlled parity, with the controls named before they ran, is the baseline a claimed quantum advantage for recorded signals has to clear.

AI USE STATEMENT

An AI assistant (Claude, Anthropic) was used to draft and refactor experiment code, to run and log the reported experiments under the authors' direction, and to edit the manuscript. All experimental design decisions, pre-registered comparisons, claims and interpretations are the authors', who verified every number against the released result files.

REPRODUCIBILITY STATEMENT

All data are public (PhysioNet). The repository contains the transform, the classical engine, every pre-registered comparison script with its decision rule, the per-window and per-patient result files, the hardware circuit constructions, the device job identifiers and raw counts, and the scripts that regenerate every table, figure and number in this paper from those files (paper/iclr2027/exp/). The three controls added in this paper run in under an hour on one consumer GPU and about 35 s of QPU time on the IBM open plan.

ETHICS STATEMENT

The study uses de-identified public EEG databases under their PhysioNet licences and involves no new human-subject data collection.

REFERENCES

- Scott Aaronson. Read the fine print. *Nature Physics*, 11:291–293, 2015.
- Armando Bellante and Stefano Zanero. Quantum matching pursuit: A quantum algorithm for sparse representations. *Physical Review A*, 105:022414, 2022.
- Armando Bellante, Stefano Vanerio, and Stefano Zanero. Quantum sparse recovery and quantum orthogonal matching pursuit. *arXiv preprint arXiv:2510.06925*, 2025.
- Joseph Bowles, Shahnawaz Ahmed, and Maria Schuld. Better than classical? The subtle art of benchmarking quantum machine learning models. *arXiv preprint arXiv:2403.07059*, 2024.
- Gilles Brassard, Peter Høyer, Michele Mosca, and Alain Tapp. Quantum amplitude amplification and estimation. In *Quantum Computation and Information*, volume 305 of *Contemporary Mathematics*, pp. 53–74. American Mathematical Society, 2002.
- Akbar Bultan. A four-parameter atomic decomposition of chirplets. *IEEE Transactions on Signal Processing*, 47(3):731–745, 1999.
- Samuel Yen-Chi Chen, Shinjae Yoo, and Yao-Lung L. Fang. Quantum long short-term memory. *arXiv preprint arXiv:2009.01783*, 2020.
- Nai-Hui Chia, András Gilyén, Tongyang Li, Han-Hsuan Lin, Ewin Tang, and Chunhao Wang. Sampling-based sublinear low-rank matrix arithmetic framework for dequantizing quantum machine learning. *Proc. 52nd ACM Symposium on Theory of Computing (STOC)*, pp. 387–400, 2020.
- Paolo Detti. Siena scalp EEG database (version 1.0.0). PhysioNet, 2020.
- Christoph Dürr and Peter Høyer. A quantum algorithm for finding the minimum. *arXiv preprint quant-ph/9607014*, 1996.
- Rosana Esteller, Javier Echaz, Thomas Tcheng, Brian Litt, and Benjamin Pless. Line length: an efficient feature for seizure onset detection. In *Proc. 23rd Annual International Conference of the IEEE EMBS*, pp. 1707–1710, 2001.
- Keisuke Fujii and Kohei Nakajima. Harnessing disordered-ensemble quantum dynamics for machine learning. *Physical Review Applied*, 8:024030, 2017.

- Ary L. Goldberger, Luis A. N. Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch. Ivanov, Roger G. Mark, Joseph E. Mietus, George B. Moody, Chung-Kang Peng, and H. Eugene Stanley. PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation*, 101(23):e215–e220, 2000.
- Alexandre Gramfort, Martin Luessi, Eric Larson, Denis A. Engemann, Daniel Strohmeier, Christian Brodbeck, Roman Goj, Mainak Jas, Teon Brooks, Lauri Parkkonen, and Matti S. Hämäläinen. MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience*, 7:267, 2013.
- Robert B. Griffiths and Chi-Sheng Niu. Semiclassical Fourier transform for quantum computation. *Physical Review Letters*, 76:3228–3231, 1996.
- Lov Grover and Terry Rudolph. Creating superpositions that correspond to efficiently integrable probability distributions. *arXiv preprint quant-ph/0208112*, 2002.
- Vojtěch Havlíček, Antonio D. Córcoles, Kristan Temme, Aram W. Harrow, Abhinav Kandala, Jerry M. Chow, and Jay M. Gambetta. Supervised learning with quantum-enhanced feature spaces. *Nature*, 567:209–212, 2019.
- Adam Holmes and A. Y. Matsuura. Efficient quantum circuits for accurate state preparation of smooth, differentiable functions. *arXiv preprint arXiv:2005.04351*, 2020.
- Hsin-Yuan Huang, Michael Broughton, Masoud Mohseni, Ryan Babbush, Sergio Boixo, Hartmut Neven, and Jarrod R. McClean. Power of data in quantum machine learning. *Nature Communications*, 12:2631, 2021.
- Hsin-Yuan Huang, Michael Broughton, Jordan Cotler, Sitan Chen, Jerry Li, Masoud Mohseni, Hartmut Neven, Ryan Babbush, Richard Kueng, John Preskill, and Jarrod R. McClean. Quantum advantage in learning from experiments. *Science*, 376:1182–1186, 2022.
- Ruiying Huang and Artur F. Izmaylov. Quantum gambling: best-arm strategies for generator selection in adaptive variational algorithms. *arXiv preprint arXiv:2509.14917*, 2025.
- Samuel Jaques and Arthur G. Rattew. QRAM: A survey and critique. *Quantum*, 9:1922, 2025.
- Ali Javadi-Abhari, Matthew Treinish, Kevin Krsulich, Christopher J. Wood, Jake Lishman, Julien Gacon, Simon Martiel, Paul D. Nation, Lev S. Bishop, Andrew W. Cross, Blake R. Johnson, and Jay M. Gambetta. Quantum computing with Qiskit. *arXiv:2405.08810*, 2024.
- S. Kannan et al. Exponential quantum advantage for learning signals with a single qubit. *arXiv preprint arXiv:2608.13521*, 2026.
- Nishant Kumar and Steve Mann. Toward a stable and deployable adaptive chirplet transform: Residual projection, hybrid GPU acceleration, and multi-channel scalability. *arXiv preprint arXiv:2607.16629 [eess.SP]*, 2026.
- Stéphane G. Mallat and Zhifeng Zhang. Matching pursuits with time-frequency dictionaries. *IEEE Transactions on Signal Processing*, 41(12):3397–3415, 1993.
- Steve Mann and Simon Haykin. The chirplet transform: A generalization of Gabor’s logon transform. In *Proc. Vision Interface*, pp. 205–212, 1991.
- Steve Mann and Simon Haykin. Adaptive “chirplet” transform: an adaptive generalization of the wavelet transform. *Optical Engineering*, 31(6):1243–1256, 1992.
- Steve Mann and Simon Haykin. The chirplet transform: physical considerations. *IEEE Transactions on Signal Processing*, 43(11):2745–2761, 1995.
- Rodrigo Martínez-Peña, Gian Luca Giorgi, Johannes Nokkala, Miguel C. Soriano, and Roberta Zambrini. Dynamical phase transitions in quantum reservoir computing. *Physical Review Letters*, 127:100502, 2021.
- Kosuke Mitarai, Makoto Negoro, Masahiro Kitagawa, and Keisuke Fujii. Quantum circuit learning. *Physical Review A*, 98:032309, 2018.

- Mudassir Moosa, Thomas W. Watts, Yiyu Chen, Abhijat Sarma, and Peter L. McMahon. Linear-depth quantum circuits for loading Fourier approximations of arbitrary functions. *Quantum Science and Technology*, 2023.
- Mikko Möttönen, Juha J. Vartiainen, Ville Bergholm, and Martti M. Salomaa. Transformation of quantum states using uniformly controlled rotations. *Quantum Information and Computation*, 5(6):467–473, 2005.
- Siyuan Niu and Aida Todri-Sanial. Enabling multi-programming mechanism for quantum computing in the NISQ era. *Quantum*, 7:925, 2023.
- Yasuhiro Ohkura, Takahiko Satoh, and Rodney Van Meter. Simultaneous execution of quantum circuits on current and near-future NISQ systems. *IEEE Transactions on Quantum Engineering*, 3, 2022.
- Y. C. Pati, R. Rezaifar, and P. S. Krishnaprasad. Orthogonal matching pursuit: recursive function approximation with applications to wavelet decomposition. In *Proc. 27th Asilomar Conference on Signals, Systems and Computers*, pp. 40–44, 1993.
- Adrián Pérez-Salinas, Alba Cervera-Lierta, Elies Gil-Fuster, and José I. Latorre. Data re-uploading for a universal quantum classifier. *Quantum*, 4:226, 2020.
- Gerwin Schalk, Dennis J. McFarland, Thilo Hinterberger, Niels Birbaumer, and Jonathan R. Wolpaw. BCI2000: A general-purpose brain-computer interface (BCI) system. *IEEE Transactions on Biomedical Engineering*, 51(6):1034–1043, 2004.
- Maria Schuld and Nathan Killoran. Is quantum advantage the right goal for quantum machine learning? *PRX Quantum*, 3:030101, 2022.
- Maria Schuld, Ville Bergholm, Christian Gogolin, Josh Izaac, and Nathan Killoran. Evaluating analytic gradients on quantum hardware. *Physical Review A*, 99:032331, 2019.
- Vivek V. Shende, Stephen S. Bullock, and Igor L. Markov. Synthesis of quantum-logic circuits. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 25(6):1000–1010, 2006.
- Ali Hossam Shoeb. *Application of machine learning to epileptic seizure onset detection and treatment*. PhD thesis, Massachusetts Institute of Technology, 2009.
- Ewin Tang. A quantum-inspired classical algorithm for recommendation systems. In *Proc. 51st ACM Symposium on Theory of Computing (STOC)*, pp. 217–228, 2019.
- Ewin Tang. Dequantizing algorithms to understand quantum advantage in machine learning. *Nature Reviews Physics*, 4:692–693, 2022.
- Alexander Vicol. Adaptive chirplet transform (ACT): software repository, v14. <https://gitlab.com/adaptive-chirplet-transform/adaptive-chirplet-transform>, 2026.
- A. Wolff, K. Hamilton, K. Rhrissorakrai, L. Parida, F. Utro, and G. Dumas. A quantum reservoir for neurodynamical forecasting. *arXiv preprint arXiv:2608.00139*, 2026.
- Igor Zyma, Sergii Tukaev, Ivan Seleznev, Ken Kiyono, Anton Popov, Mariia Chernykh, and Oleksii Shpenkov. Electroencephalograms during mental arithmetic task performance. *Data*, 4(1):14, 2019.

A PROOFS

Proposition 1. With $|s\rangle = \sum_t s(t)|t\rangle$ and the conjugate chirp $D_c|t\rangle = e^{-i2\pi ct^2/N}|t\rangle$, the inverse QFT maps $|t\rangle \mapsto N^{-1/2} \sum_k e^{-i2\pi kt/N}|k\rangle$, so the amplitude on $|k\rangle$ is $N^{-1/2} \sum_t s(t)e^{-i2\pi ct^2/N}e^{-i2\pi kt/N}$ and eq. equation 3 follows. Writing $\psi_{e,c,k} = e \odot \chi_{c,k}/\|e\|$ with $\chi_{c,k}(t) = e^{i2\pi(ct^2+kt)/N}$ and using $s = e \odot r/\|e \odot r\|$, $\langle \psi_{e,c,k} | r \rangle = \|e\|^{-1} \sum_t e(t)r(t)\chi_{c,k}(t) = \|e \odot r\| \|e\|^{-1} \sum_t s(t)\chi_{c,k}(t)$, whose squared modulus is $N\|e \odot r\|^2 \|e\|^{-2} P_{e,c}(k)$; dividing by $\|\psi\|^2 = 1$ gives the claim, and the sum over t for all k is one discrete Fourier transform of $e \odot r \odot e^{-i2\pi ct^2/N}$.

Proposition 2. R_k is unitary and Hermitian with $(I + R_k)/2 = I - |\psi_k\rangle\langle\psi_k| = P_k$. On $|0\rangle|r\rangle$, the first Hadamard gives $(|0\rangle + |1\rangle)|r\rangle/\sqrt{2}$, the controlled reflection gives $(|0\rangle|r\rangle + |1\rangle R_k|r\rangle)/\sqrt{2}$, and the second Hadamard gives $|0\rangle(I + R_k)|r\rangle/2 + |1\rangle(I - R_k)|r\rangle/2$; the ancilla reads 0 with probability $\|P_k r\|^2/\|r\|^2$ and leaves $P_k|r\rangle/\|P_k r\|$. Chaining, $P_K \cdots P_1|x\rangle$ is the matching-pursuit residual r_K because $r_k = P_k r_{k-1}$ by definition, and the success probabilities multiply to $\prod_k \|r_k\|^2/\|r_{k-1}\|^2 = \|r_K\|^2/\|x\|^2$. The controlled reflection needs only the reflection about $|0\rangle$ to be controlled: $c-R_k = (I \otimes U_k) c - (I - 2|0\rangle\langle 0|)(I \otimes U_k^\dagger)$, a multi-controlled phase of π conjugated by $X^{\otimes n}$.

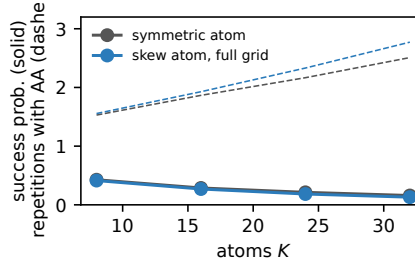


Figure 3: Coherent-residual success probability (solid) and repetitions under amplitude amplification (dashed) as a function of the number of atoms, from the mean squared relative residual of the classical study on 320 public EEG windows (Vicol, 2026).

Complexity of the superposed search. With the triples $(t_c, \Delta t, c)$ in index registers, the envelope preparation and the phase gates of eq. equation 2 become controlled on those registers; the outcome register holds k . Quantum maximum finding (Dürr & Høyer, 1996) over (index, k) with the amplitude estimated to precision ε (Brassard et al., 2002) costs $O(\sqrt{M'N_f}\varepsilon^{-1})$ preparations of the searched state, each of which loads x ($\Theta(N)$ gates without QRAM) and applies the $k - 1$ reflections of Proposition 2 ($O(kn^2)$ plus envelope preparations), giving the bound quoted in §4. The N_f factor is absorbed into M' in the text. With coherent access to the signal (QRAM, or a signal that reaches a qubit directly, Huang et al., 2022; Kannan et al., 2026), the same construction runs in $O(\sqrt{M'}\varepsilon^{-1}(Kn^2 + \text{polylog } N))$, which is the regime the quantum matching-pursuit literature analyses.

B REFINEMENT CONTROL: ADDITIONAL DETAIL

Table 5 gives the backfit and extended-band arms; the extended arm is the shipped grid extended to 78 Hz (39,780 atoms, \$2). Windows on which the classical engine as shipped leaves more than 90% of the energy have 84% of their energy above 45 Hz (mains); the others 11%. With the backfit pass QACT parity has, the classical engine on the extended grid is -14.2% relative to QACT parity at order 12 (better on 285 of 320, $p = 1 \times 10^{-38}$).

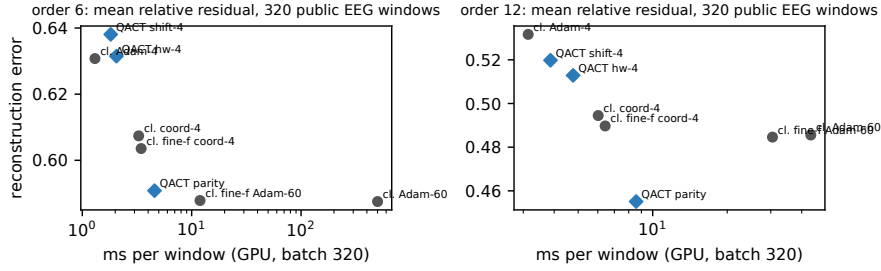


Figure 4: Reconstruction error against GPU time per window for the engines of Table 3 (grey: classical; blue: QACT). At equal refinement rule the classical engine is ahead; the QACT parity mean at order 12 is carried by the windows on which the shipped grid fails (§5.2).

Table 5: Backfit and extended-band arms, same windows as Table 3: mean and median relative residual at order 12, change versus the classical engine as shipped, windows better, paired p .

engine	mean@6	mean@12	median@12	vs ref@12	better / p
classical, shipped grid, Adam 60 (reference)	0.588	0.486	0.437	—	
classical, shipped grid + backfit 1	0.580	0.479	0.430	-1.3%	244 / 2×10^{-16}
classical, shipped grid to 78 Hz	0.512	0.398	0.387	-18.1%	125 / 1×10^{-22}
classical, shipped grid to 78 Hz + backfit 1	0.505	0.391	0.379	-19.6%	278 / 5×10^{-36}
QACT parity	0.591	0.455	0.447	-6.3%	104 / 0.05

C DOWNSTREAM COMPARISONS: FULL TABLES

Seizure detection. CHB-MIT: 10,856 two-second epochs (5,501 ictal / 5,355 interictal), 23 patients, cross-patient folds. With all 270 chirplet + amplitude features and regularisation (no selection), the gain of chirplet features over amplitude-only, paired across patients (decision threshold $p < 0.0036$): logistic regression $C = 0.01$, raw, +8.6 points ($p = 0.0011$, 20/23); $C = 0.1$, raw, +7.5 ($p = 0.0005$, 20/23); RBF SVM, raw, +3.4 ($p = 0.050$); logistic $C = 0.01$, per-patient normalised, +3.5 ($p = 0.0096$); $C = 0.1$, normalised, +3.4 ($p = 0.0015$, 17/23); RBF SVM, normalised, +1.9 ($p = 0.039$). Best absolute accuracies (an untuned RBF SVM on all features): 81.4% on CHB-MIT with per-patient normalised features, 76.8% raw, and 79.2% on Siena. Under univariate top- k selection every model selected all 36 amplitude features first and the “chirplet + amplitude” result was in fact amplitude-only, which is why the selection regime was replaced. QACT versus classical features, all features and regularisation: logistic L2 62.7 vs 63.6%, L1 62.9 vs 63.4%, RBF SVM 66.4 vs 67.5%; sampled versus argmax selection for the SVM -1.93 points, $p = 0.0032$. Siena (14 patients, 45 seizures, never inspected during development): amplitude only 69.5 / 65.9% (logistic / SVM), chirplet + amplitude 78.9 / 79.2%, QACT + amplitude 78.5 / 79.1%; chirplet gain +9.43 ($p = 0.0047$) and +13.31 ($p = 0.0040$); QACT versus classical +0.30 and -1.99, not significant.

Atom quality versus feature value. Atom-level improvements that are real (the refiner, cubic and skew atoms, measurement as a proposal distribution that halves the error on planted chirplets) produced no downstream gain. The best of them had the best atoms and the worst features, and on real EEG its reconstruction advantage did not exist (0.3797 versus 0.3828), because 1/f EEG has no clean atoms to find.

Quantum classifiers. CHB-MIT, ACT + amplitude features, pre-registered win = best quantum > best classical and $p < 0.025$: RBF SVM 67.7% raw / 79.2% per-patient normalised; projected quantum kernel 66.9 / 78.0; QSVC 65.8 / 78.2; quantum-feature logistic regression 63.8 / 78.7; hybrid 63.9 / 76.8 and its no-entanglement twin 63.9 / 76.8; logistic regression 64.5 / 76.1. EEGMAT (rest versus arithmetic, 36 subjects, nested subject-wise CV, equal tuning budgets): logistic regression 55.0 / 57.4% (raw / per-subject normalised), RBF SVM 51.6 / 55.7, QSVC 53.5 / 54.4, projected kernel 52.7 / 54.3; quantum features with linear readout 52.8 / 56.6 against 54.5 / 57.9 classical; the entangled hybrid 53.1 / 56.5 against its non-entangled twin 54.8 / 57.9. The alpha-suppression effect

that validates the loading is present at the recording level (-25.5% , 8/8 subjects, $p = 0.0078$) and nearly washes out per epoch.

Forecasting. Per-person models on the first half of each CHB-MIT recording, tested on the second half (490 h each), 270 features in 10 s steps, targets the next minute of features and a seizure-within-5-minutes probability. LSTM skill $+0.106$; fixed quantum-circuit features $+0.101$ against twin $+0.102$; quantum LSTM $+0.014$ against twin $+0.019$. Quantum reservoirs (four parallel 8-qubit disordered transverse-field Ising systems, input-qubit replacement, ridge readout; exact density-matrix simulation verified against a circuit simulator to 10^{-7}): on NARMA-10 and memory capacity the quantum reservoir beats an echo-state network of the same readout size (0.191 vs 0.240 error; capacity 14.8 vs 11.5); on EEG at 2 s steps run continuously, ESN $+0.436$, quantum $+0.429$, zero-coupling twin $+0.405$; the entanglement contribution grows monotonically as the step shortens (60 s: -0.004 , $p = 0.23$; 2 s: $+0.024$, 23/23, $p < 10^{-4}$) and the classical reservoir wins at every step size.

D THE STUDY’S LATER SEARCH FOR A QUANTUM WIN

Summary. A hybrid transform routed by register size keeps classical quality up to seven qubits and is slower at every level; error correction restores long-atom quality at 6,000–310,000 physical qubits and 12 minutes to 111 hours per window; a pre-registered 60-window test finds the eight-qubit hybrid indistinguishable from classical and from a noise-mixed classical control; fault-tolerant coherent search crosses over with FFT matching pursuit only at 10^{13} atoms; kernels on atoms and small-data regimes land at parity or below; and entangled sensing probes gain nothing at realistic spin numbers, because the brain’s own background sets the trial count.

Detail. All numbers are from the study’s result files (`results/hybridact_levels.json`, `qact_qec.json`, `q8_vs_classical.json`, `qsearch_*.json`, `quantum_sensing_opm.json`) and its search log (`docs/QACT_SEARCH_LOG.md`). *Hybrid levels* (4 EEGMAT windows, 6 atoms, quantum routes executed under the Heron noise model): fit error 0.578 classical; 0.575, 0.567, 0.561 at levels q6–q8 (2–21% of evaluations quantum, chosen atoms at 99–91% of the best on average, worst 80–54%); 0.734 at q9 (all quantum, worst atom 31%); every level slower than the 45 ms classical run. *Error correction* (Microsoft resource estimator, surface code, logical-level runs in Aer): Heron as calibrated is above threshold on readout error; with 10^{-3} or 10^{-4} physical error rates q8/q9 recover their noiseless quality (0.573/0.598) at 6,480–95,256 physical qubits and 12 minutes to 4.8 hours per window; with perfect qubits q9 remains slightly worse than classical (0.598 vs 0.578) from shot noise and coarser window grids. *Powered test* (60 fresh windows, 30 subjects, paired Wilcoxon, $\alpha = 0.0125$): q8 on Heron $+0.0007$ ($p = 0.79$), error-corrected $+0.0016$ ($p = 0.077$), classical noise-mixed controls -0.0022 and $+0.0008$, noiseless q8 $+0.0063$ ($p = 0.0014$, worse). *Fault-tolerant coherent search*: Dürr–Høyer with amplitude estimation, every constant tilted toward quantum and QRAM free, crosses over with classical FFT matching pursuit at 1.5×10^{13} atoms. *Learners on chirplet atoms* (CHB-MIT, patient-wise): a quantum kernel on sets of atoms 62.6% vs 67.4% RBF-SVM ($p = 0.0015$), an all-feature kernel 74.8% vs 76.8% ($p = 0.09$), learning curves from 20 to 200 training epochs at parity once kernel concentration was controlled. *Sensing*: on real OPM-MEG evoked fields a noiseless sensor needs only $1.1\text{--}2.3\times$ fewer trials than today’s OPM, and entangled probes gain $\geq 10\%$ only below 3×10^9 spins with $T_2 \geq 1$ s.

E HARDWARE DETAILS

Circuit counts. Per atom, branch-and-bound over envelopes with the classical bound $(\sum e|r|)^2/\|e\|^2$ visits 57% of the circuits with identical selections, and the three-point refiner replaces 112 evaluations per step by 8: about $80\times$ fewer circuit executions per atom.

Throughput techniques. The three techniques behind the 8.0 s estimate are zero repetition delay; multi-programming 14 circuits per chip (Niu & Todri-Sanial, 2023; Ohkura et al., 2022); and adaptive shot allocation by successive elimination (Huang & Izmaylov, 2025), run here and cutting shots $2.9\times$ at identical error.

Device against noise model. Measured Hellinger fidelities (0.76 and 0.62 for six and seven qubits) sit below the fake-backend predictions (0.92 and 0.81), so the calibrated model is optimistic by about 0.15–0.2 on this device today. The unitary-QFT form with twirling (0.70 mean fidelity) held up slightly better than the dynamic form without it (0.68).

Table 6: Time per window for the same decomposition job. Device rows price the circuits the simulator run actually executed, at the device’s calibrated gate and measurement durations; queueing and compilation excluded.

engine	per window	vs classical CPU	recon. error
classical engine, GPU, batch 1,024	0.672 ms	82× faster	0.47
classical engine, CPU	55.38 ms	–	0.47
QACT exact simulation, GPU, batch 1,024 (the dequantised program)	0.088 ms	630× faster	0.58
QACT circuits shot by shot, simulator (1,140 circuits)	54 s	971× slower	0.58
IBM Heron, optimised circuits (estimate)	758 s	13,700× slower	
+ zero repetition delay, multi-programming, adaptive shots	8.0 s	144× slower	
IBM Heron, original design (estimate)	17.0 h	$1.1 \times 10^6 \times$ slower	

Circuits are built from Qiskit library components (`StatePreparation`, `QFTGate`, `UCRYGate`) and transpiled with the preset pass manager at optimisation level 3 to the device target (heavy-hex coupling, native CZ, RZ, SX, X, mid-circuit measurement and feed-forward). Gate counts in Fig. 1 are from a Heron target; noisy simulation uses that backend’s calibrated noise model. The device run used `ibm_marrakesh` through the open plan with dynamical decoupling (XY4) on both jobs and gate and measurement twirling (16 randomisations) on the unitary-QFT job; the dynamic-circuit job could not be twirled. The exact target for every circuit, its transpiled gate counts, the fake-backend prediction, the device counts and the job identifiers are in `paper/iclr2027/results/`.